

# Internship Report – Day 6

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Internship Program: ParkNSecure – Smart Parking Solutions

## Day 6 Overview:

Today we enhanced the ANPR system by implementing an authorization system that checks detected number plates against a database of authorized Indian vehicles. We created a CSV file containing 10 authorized Indian number plates (following the standard format of state code, district number, series, and unique number) along with owner details and access types. The system now provides real-time feedback by displaying green rectangles for authorized vehicles and red for unauthorized ones, while printing detailed information in the terminal including the plate number, authorization status, owner name, and access type. The improvements make the system more suitable for Indian vehicle number plate recognition and access control applications.

## Tools Used Today:

VS CODER, GEMINI, SONNET.

## Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

## Approach to the Task:

The ANPR system implements a comprehensive approach combining data management through a structured CSV file for authorized Indian number plates, real-time detection using EasyOCR and OpenCV for text recognition and video processing, an efficient authorization system using dictionary-based lookups with color-coded

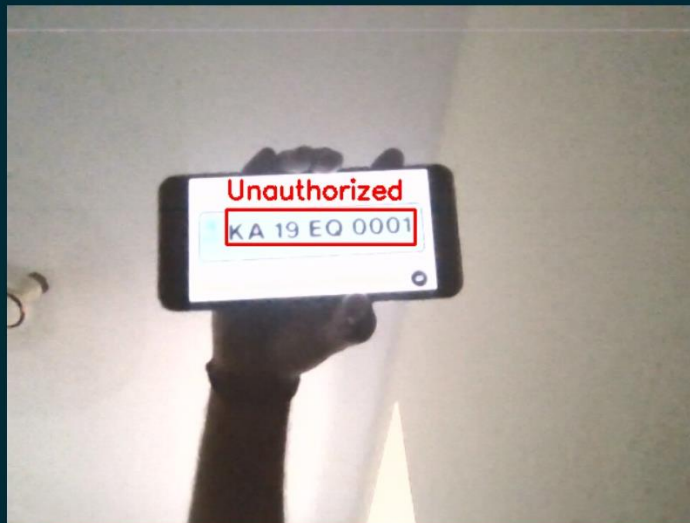
visual feedback (green for authorized, red for unauthorized), and a user-friendly interface with clear terminal messages and visual indicators. The system processes each frame by cleaning detected text (removing spaces, converting to uppercase), checking against the authorized database, and providing immediate feedback through both visual indicators and detailed terminal output, while automatically saving images of detected plates and handling errors gracefully.

## Attachments:-

```
37     ret, frame = cap.read()
38     if not ret:
39         break
40
41     # Optional: Resize for faster processing
42     img = cv2.resize(frame, (640, 480))
43
44     # Run OCR on the frame
45     results = reader.readtext(img)
46     detected = False
47     for (bbox, text, prob) in results:
48         # Draw rectangle around detected text
49         (top_left, top_right, bottom_right, bottom_left) = bbox
50         top_left = tuple(map(int, top_left))
51         bottom_right = tuple(map(int, bottom_right))
52
53         # Clean the detected text (remove spaces and convert to uppercase)
54         cleaned_text = text.replace(" ", "").upper()
55
56         # Check if the plate is authorized
57         if cleaned_text in authorized_plates:
58             # Authorized plate - draw green rectangle
59             cv2.rectangle(img, top_left, bottom_right, (0, 255, 0), 2)
60             plate_info = authorized_plates[cleaned_text]
61             status_text = f'Authorized: {plate_info["owner_name"]}'
62             print(f'Number Plate Detected: {cleaned_text}')
63             print(f'Status: AUTHORIZED')
64             print(f'Owner: {plate_info["owner_name"]}')
65             print(f'Access Type: {plate_info["access_type"]}')
66         else:
67             # Unauthorized plate - draw red rectangle
68             cv2.rectangle(img, top_left, bottom_right, (0, 0, 255), 2)
69             status_text = "Unauthorized"
70             print(f'Number Plate Detected: {cleaned_text}')
71             print(f'Status: UNAUTHORIZED')
72
73         # Show status text
74         cv2.putText(img, status_text, (top_left[0], top_left[1] - 10),
75                     | | cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0) if cleaned_text in authorized_plates else (0, 0, 255), 2)
76         detected = True
77
78     if detected:
79         timestamp = datetime.now().strftime('%Y%m%d_%H%M%S_%f')
80         filename = f'frame_{timestamp}.jpg'
81         filepath = os.path.join(output_dir, filename)
82         cv2.imwrite(filepath, img)
83         print(f'Saved frame with detection to: {filepath}')
84         frame_count += 1
85     except KeyboardInterrupt:
86         print("\nStopped by user.")
87
88     cap.release()
```

```
authorized_plates.csv > data
1  plate_number,owner_name,access_type
2  KA01AB1234,Rahul Sharma,Full Access
3  DL01CD5678,Priya Patel,Full Access
4  MH02EF9012,Amit Singh,Full Access
5  TN03GH3456,Neha Gupta,Full Access
6  KL04IJ7890,Rajesh Kumar,Full Access
7  KA05KL2345,Anita Desai,Full Access
8  DL06MN6789,Vikram Malhotra,Full Access
9  MH07OP0123,Deepika Reddy,Full Access
10 TN08QR4567,Arun Verma,Full Access
11 KL09ST8901,Meera Joshi,Full Access |
```

```
Number Plate Detected: KA19EQ0001
Status: UNAUTHORIZED
Saved frame with detection to: anpr_output\frame_20250510_174846_612347.jpg
```



Problems 3 Output Debug Console Terminal Ports powershell + - [ ] [ ] ... ^ x

```
Number Plate Detected: KA19EQ0001
Status: UNAUTHORIZED
Saved frame with detection to: anpr_output\frame_20250510_174845_791912.jpg

Number Plate Detected: KA19EQ0001
Status: UNAUTHORIZED
Saved frame with detection to: anpr_output\frame_20250510_174846_612347.jpg
```